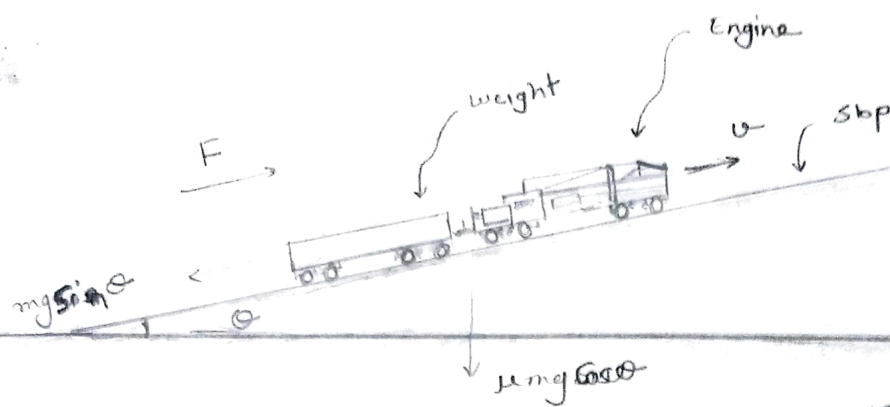


age Using trains:-



F_r = force Required to move the Train

$$F_r = mg \sin \theta + \mu mg \cos \theta$$

$$F_r = mg (\sin \theta + \mu \cos \theta)$$

where

$m \rightarrow$ mass

$\mu \rightarrow$ coefficient of friction

$g \rightarrow$ acceleration due to gravity.

$F \rightarrow$ minimum Required force to Run the Train.

Torque Required for this force

$$T = \frac{F_r \times \text{Radius of the wheel (r)}}{\text{Gear Ratio (Gr)}}$$

$$T = \frac{mg (\sin \theta + \mu \cos \theta) \times r}{Gr}$$

Speed of the Train

$$v = \frac{2\pi r \times \text{RPM}}{60}$$

angular velocity

$$\omega = \frac{\text{Power}}{\text{Torque}}$$

where the RPM of the motor/generator inside the train must Synchronize while discharging the Battery

Then we Need Synchronous Speed while discharging

$$\text{Synchronous Speed (Hz)} = \frac{120 f}{P}$$

By using this formula we can easily find the Speed of train while discharging.

The potential energy stored by the train :-

$$E = mgd \cdot \sin \theta \text{ (J)}$$

where
d is displacement of the train travelled on the incline surface.

Conversion of mechanical potential energy stored by the train to the electrical energy :-

The conversion of mechanical energy using motor generally depends on the efficiency of generator

$$E_{\text{electrical}} = P \cdot E_{\text{mechanical}} \times \eta \text{ (J)}$$

where this energy is not the actual energy converted in electrical energy.

$$E_{\text{electrical}} = mgd \sin \theta \cdot \eta_{\text{generator}} \text{ (J)}$$

→ This is the only the electrical energy stored in the plant

Generation of Electricity by run the train on slope freely :-

When train start moving the potential energy is converted to kinetic energy.

$$KE = \frac{1}{2} m v^2$$

friction forces that oppose the motion

$$f_{\text{friction}} = \mu \cdot mg \cdot \cos \theta$$

Work done to overcome friction

$$W_{\text{friction}} = f_{\text{friction}} \cdot L$$

where f_{friction} & L is the length of the track.

Net energy available for electricity generation

$$E_{\text{net}} = PE - W_{\text{friction}}$$

Energy Conversion by Generators :-

$$E_{\text{elec}} = E_{\text{net}} \cdot \eta$$

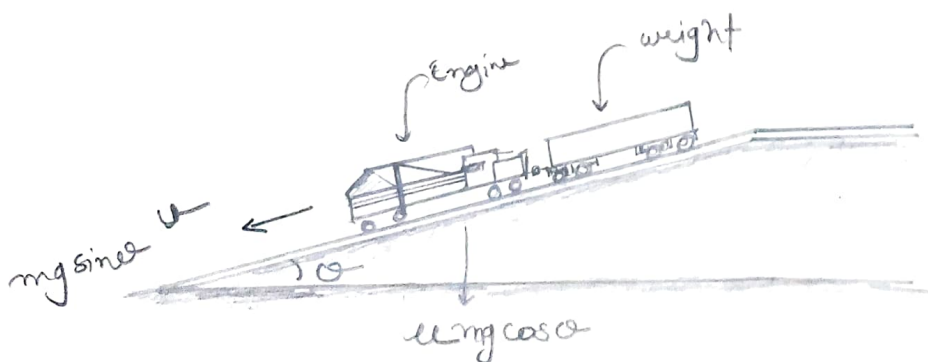
By Balancing the Kinetic energy of train and the ~~net~~ E_{net} we can also find the speed of the train.

$$E_{\text{elec}} = m g d \sin \theta - \mu m g \cos \theta \cdot L$$

$$E_{\text{elec}} = \eta_0 \cdot m g (d \sin \theta - \cos \theta \cdot L)$$

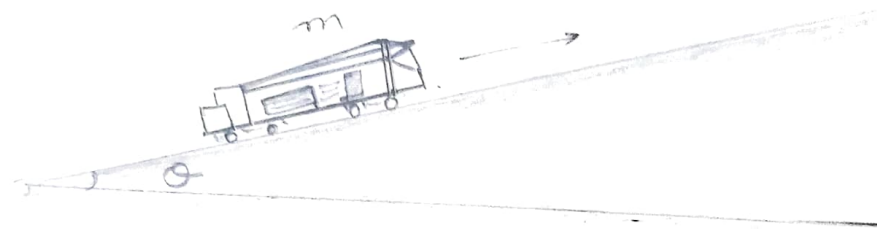
efficiency of generator.

This electric energy is also not the complete energy that we get because we do include friction loss by air and other constant loss.



The advantages of Train energy storage system :-

(1)



Suppose we have an engine of mass m running on u velocity
Then the force and torque required to run the engine

$$F = mg(\sin\theta + \mu\cos\theta)$$

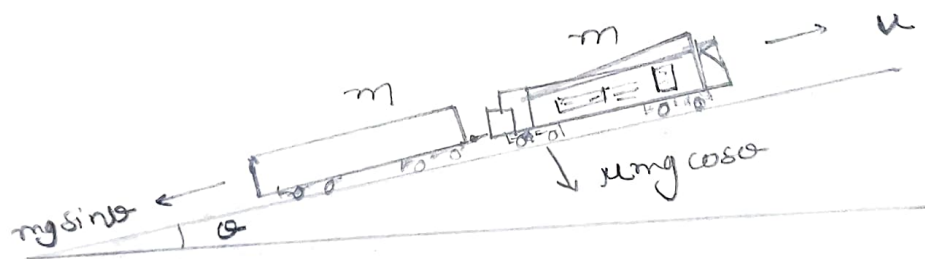
$$T = \frac{F \times r}{Grr}$$

where

$r \rightarrow$ Radius of wheel

$Grr \rightarrow$ Gear Ratio

If we want to ~~store more~~ store electricity then we add another mass of whi with train.



Then we only have to do this half the gear ratio
 $F = 2mg(\sin\theta + \mu\cos\theta)$

$$T = \frac{F \times r}{Grr}$$

$$T = \frac{F \times r}{\left(\frac{Grr}{2}\right)}$$

$$T = \frac{2 F \times r}{Grr}$$

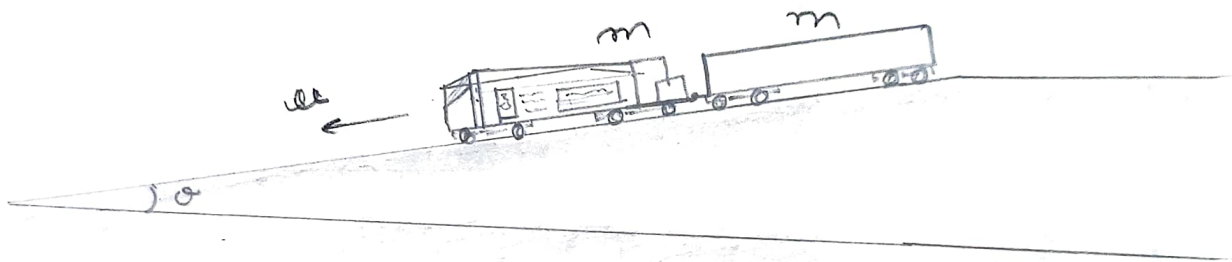
Speed.
obioslly the $\frac{1}{T}$ will decrease then

$$\boxed{\frac{1}{T} \propto u}$$

Conclusion :- we can store more power on the same track by adding mass and by changing the gear Ratio.

Note :- We can only add the weight till the $mg \sin \theta < \mu mg \cos \theta$

(2) Suppose the train coming towards downhill

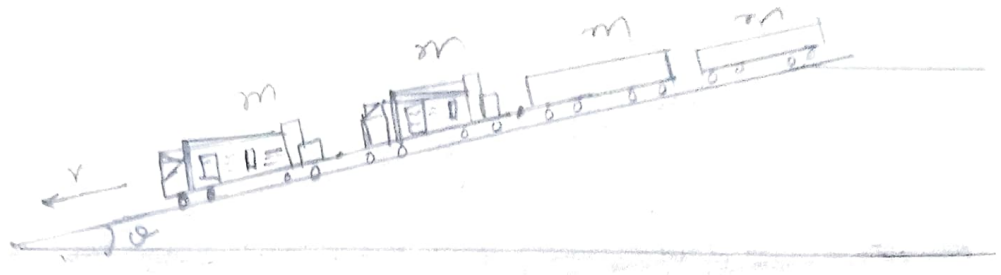


Then the equations for that

$$F = 2mg (\sin \theta + \mu \cos \theta)$$

$$T = 2 \frac{F \times r}{\text{Cirr}}$$

Suppose there are the peak load time the the electric load on the grid is jump suddenly the we generate double the power by adding an extra engine and an extra mass without changing its velocity and mass



Conclusion:- we can make energy according to our demand
 It helps to stabilize the grid and prevent
 from short term power cutoffs.

Note:- There are some limitations to increase the
 weight on this system.

Overall conclusion:- we can store more electricity
 on the same track by changing
 the weight and torque and we can also generate
 more electricity by adding engines and weights.

Example 8-

Mass of the Train = 10,000 kg

Track length = 1000 m

Incline angle $\theta = 2.87^\circ$

Coefficient of friction $\mu = 0.01$

Gravitational acceleration $g = 9.8 \text{ m/s}^2$

Efficiency of the generator $\eta = 70\%$

Calculating the mechanical energy stored:

$$P.E = m g d \sin \theta$$

$$P.E = 4,905,000 \text{ J}$$

friction losses:

$$f_{\text{friction}} = \mu m g \cos \theta$$

$$f_{\text{friction}} = 0.01 \cdot 10000 \cdot 9.8 \cos(2.87^\circ)$$

$$f_{\text{friction}} = 980 \text{ N}$$

$$W_f = 980000$$

$$E_{\text{Net}} = P.E - W_{\text{friction}}$$

$$E_{\text{Net}} = 5,607,143 \text{ J}$$

electrical energy required to generate this mechanical energy is 186.9 kW (when the $t = 30 \text{ s}$)
Time

$$v = \frac{E}{t} = \frac{50.07}{30 \text{ s}}$$

Speed of the train $v = 1.67 \text{ m/s}$

Torque Required

$$F_{\text{eff}} = m \cdot g \cdot \sin(\theta) + \text{friction}$$

$$F_{\text{eff}} = 5,888 \text{ N}$$

$$T = 2,944 \text{ Nm} \quad (\text{where } r = 0.5 \text{ m})$$

→ radius of the wheel

RPM of the motor.

$$\omega = \frac{v}{r}$$

$$\omega = \frac{1.67}{0.5}$$

$$\omega = 3348 \text{ rad/s}$$

$$\text{RPM} = \frac{\omega \cdot 60}{2\pi}$$

$$\text{RPM} = 31.9$$

$$\text{Gear ratio} = \frac{\text{Motor RPM}}{\text{Drum RPM}}$$

while
RPM

$$G_r = 94$$

Electrical energy generated.

$$E_{\text{el}} = E_{\text{net}} \cdot \eta$$

$$E_{\text{el}} = 3,923,143 \text{ J} \times 0.70$$

$$E_{\text{el}} = 2,747,500 \text{ J}$$

$$E_{\text{el}} = 91,583 \text{ W}$$

$$E_{\text{el}} = 91 \text{ kW}$$

Then the efficiency of whole system is nearly 50%.

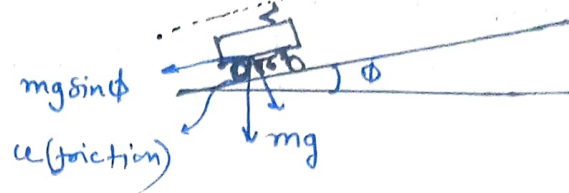
→ 50%

Note: The efficiency of the whole system is 50%. when the motor is 70% efficient but the efficiency of the motor will increase or in between 80-85%.

force required to move the train =

friction force + $mg \sin \phi$

$$F_{\text{Total}} = m \cdot g \cdot (\sin \theta + \mu \cos \theta)$$



where $f_{\text{friction}} = \mu mg \cos \phi$

The energy stored :-

$$E = mgh$$

$$E = mgd \sin \phi \quad (\text{where } d \text{ is the distance})$$

for Example :-

$$\eta = 1$$

$$m = 10000 \text{ kg}$$

$$d = 1000 \text{ m}$$

$$\mu = 0.01$$

Then

~~the~~

$$\text{angle} = 2.87^\circ$$

$$E_{\text{potential}} = 4,900,000 \text{ J.}$$

for that case :-

$$v = \frac{2\pi \times 0.5 \times 1500}{60}$$

$$v = 7.8 \text{ m/s}$$

$$T_w = \frac{5878 \times 0.5}{10}$$

Torque Required :-

$$T = \frac{F \times r}{\text{Gear Ratio.}}$$

$$\text{Speed } \omega = \frac{2\pi r \times \text{RPM}}{60}$$

(where r is the radius of the wheel)

$$T_w = 293.9 \text{ Nm}$$

$$T_m = T_w \times \text{Gear Ratio.}$$

$$T_m = 2,939 \text{ Nm}$$